

C16QS LTE Modem Baseboard – Design Note

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EDA Tool: KiCad

PCB Stackup: 2-Layer (Top Signal, Bottom Ground Plane)

1. Design Overview

This project implements a compact LTE Cat-1 modem baseboard using the Cavli C16QS module. The design focuses on meeting all mandatory Cavli hardware requirements while maintaining a minimal BOM, clear signal routing, and manufacturability suitable for prototyping and small-scale production. The board integrates power regulation, UART communication, SIM interface, RF connectivity, and boot/reset control.

2. Power Supply Design and Budget

The C16QS requires a VBAT supply between 3.1 V and 4.5 V, with a typical operating voltage of 3.8 V and peak current consumption of approximately 830 mA during LTE transmission. A switching buck regulator (XL1509-ADJ) is used to convert a 5 V input to a regulated 3.8 V rail (VBAT_3V8). The regulator provides adequate current headroom to handle LTE current bursts. Bulk and high-frequency decoupling capacitors are placed close to the regulator and modem power pins to ensure voltage stability and prevent brownout during RF activity.

3. UART Interface and Level Shifting

The C16QS uses 1.8 V digital I/O, while external host interfaces typically operate at 3.3 V. A bidirectional level shifter is used on the UART1 TX and RX lines to ensure safe and reliable communication. The modem's VDD_EXT (1.8 V) output is used to power the low-voltage side of the level shifter, while the high-voltage side interfaces with a 3.3 V UART header for AT command communication and debugging.

4. Boot and Reset Control

Auto-boot functionality is implemented by pulling the PWRKEY_N signal to VDD_EXT through a resistor, allowing the modem to power on automatically when VBAT is applied. The reset pin (RSTN) is connected to a small capacitor to ground to improve noise immunity.

and prevent unintended resets caused by power or RF transients. Test access is provided for recovery and debugging.

5. SIM Card Interface

A nano-SIM socket is connected to the modem's USIM interface. SIM power is provided directly by the modem through USIM_VDD, supporting both 1.8 V and 3.0 V SIM cards. Series resistors are placed on SIM CLK, DATA, and RST lines to improve signal integrity and provide basic ESD protection. A SIM presence detect signal is routed to enable card detection.

6. RF and Antenna Implementation

The LTE RF path is routed from the ANT_LTE pin through a π -matching network footprint to a u.FL antenna connector. The matching components are marked as DNP to allow antenna tuning during validation. The RF trace is kept short and straight on the top layer and referenced to a continuous bottom-layer ground plane with stitching vias to maintain impedance control and reduce EMI.

7. PCB Layout and DRC Status

The PCB is implemented as a 2-layer design with a solid bottom ground plane to support RF performance and power integrity. Component placement prioritizes short power paths and isolation between RF, switching power, and digital sections. During final verification, the PCB Design Rule Check reports minor clearance warnings related to compact routing. These are within standard fabrication tolerances for common 2-layer PCB processes and do not affect electrical functionality or manufacturability. Some DRC errors such as shorts or unconnected nets are present.